Senthamizhan Rajendran

Nonlinear Dynamics · Synchronization · Collective Dynamics of Complex Systems
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Education

Ph.D. in Physics, SASTRA Deemed University, Thanjavur 2022 – 2026
Thesis: *Theoretical, Computational, and Data-driven Analysis of collective dynamics in Kuramoto and swarmalator models*. viva voce completed on 26/08/26.

M.Sc. in Physics, SASTRA Deemed University, Thanjavur 2019 – 2021

B.Sc. Physics & Computer Science (dual major), SASTRA Deemed University 2016 – 2019
Srinivasa Ramanujan Centre, Kumbakonam.

Publications

Peer-reviewed journal articles

1. **R. Senthamizhan**, R. Gopal and V. K. Chandrasekar. Swarmalators with frequency-weighted interactions. *Physical Review E* **113**, 034216 (2026). doi:10.1103/dhz7-p4yg
2. **R. Senthamizhan**, R. Gopal and V. K. Chandrasekar. Frustration-induced chimeras and motion in two-dimensional swarmalators. *Chaos, Solitons & Fractals* **194**, 116164 (2025). doi:10.1016/j.chaos.2025.116164
3. **R. Senthamizhan**, R. Gopal and V. K. Chandrasekar. Data-driven exploration of swarmalators with second-order harmonics. *Physical Review E* **109**, 064303 (2024). doi:10.1103/PhysRevE.109.064303

Under review

4. **R. Senthamizhan**, R. Gopal and V. K. Chandrasekar. Effect of biharmonic coupling in frequency-weighted Kuramoto oscillators. Under review at *Chaos: An Interdisciplinary Journal of Nonlinear Science* (submitted 30 July 2026).

Research Experience

Doctoral Researcher · Department of Physics, SASTRA Deemed University 2022 – 2026

– Derived the analytical conditions separating collective states in Kuramoto and swarmalator systems using mean-field theory, self-consistency arguments and linear stability analysis, confirming every predicted boundary against direct numerical simulation.
– Established that second-order harmonics, phase frustration and frequency-weighted coupling each produce distinct clustered, chimera, translational and mixed states.
– Built the simulation and analysis pipeline behind four first-author papers: Fortran 95 RK4 integrators for large-population runs, Numba-accelerated Python for parameter sweeps, and GPU execution on HPC clusters.
– Trained a U-Net (TensorFlow/Keras) on multichannel statistical maps of simulated swarm dynamics to segment the parameter plane and locate an analytically intractable state boundary, extracted by Sobel edge detection and checked against order parameters.

Project Associate · SASTRA Deemed University Aug 2024 – Jul 2026

– Analytical modelling and scalable Python simulation workflow on a sponsored nonlinear-dynamics project. Funded by the DST-SERB Core Research Grant No. CRG/2023/003505 (PI: Dr. R. Gopal).

Teaching

Teaching Assistant · Department of Physics, SASTRA Deemed University 2022 – 2024

Awards

– **Best Poster Award** : International Conference on Nonlinear Systems and Dynamics (CNSD 2026), IISER Thiruvananthapuram, July 2026.
– **Best Poster Award** : International Conference on Complexity and Nonlinear Dynamics in STEM (CNLDS 2023), IIT Hyderabad, June 2023.

Conference Presentations

– *Swarmalators with frequency-weighted interactions.* Poster, CNSD 2026, IISER Thiruvananthapuram, July 2026.
– *Influence of second-order harmonics in two-dimensional swarmalators.* Poster, CNLDS 2023, IIT Hyderabad, June 2023.

Technical Skills

Analytical	Mean-field theory, self-consistency analysis, linear stability analysis
Languages	Python, Fortran 95, Latex
Computation	Numerical integration (RK4 and multiple ODE solvers), Numba (njit), NumPy, Matplotlib, HPC cluster workflows
Machine learning	TensorFlow/Keras (U-Net, CNNs, RNN/LSTM); PyTorch and CUDA for GPU-accelerated simulation
Tools	Linux CLI, Git, shell scripting

References

Dr. R. Gopal
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Assistant Professor, SASTRA Deemed University
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Dr. V. K. Chandrasekar
Co-supervisor
Associate Professor, SASTRA Deemed University
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